

Diffusion in Generalized Hydrodynamics

Reconstructing the bridge from current fluctuations to diffusion

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Advisor discussion: a guided refresher

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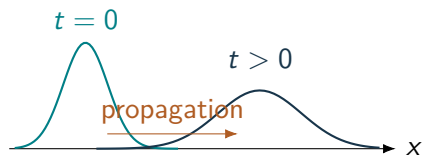
The question: how does a conserved disturbance move and broaden?

- ▶ Euler GHD: state-dependent propagation.
- ▶ Next order: additional diffusive broadening.
- ▶ Interactions already enter Euler velocities; diffusion is not synonymous with quasiparticle decay.

$$\omega(k) = vk - \frac{i\gamma}{2}k^2 + \dots$$

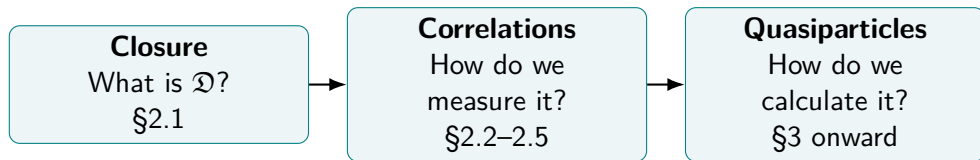
Centre $\sim vt$; additional variance $\sim \gamma t$.

Source: Introduction; scalar illustration in the study notes.



Schematic single-mode packet, not calculated model data. A many-charge profile can also split ballistically.

The route: define, identify, then calculate



$$\mathfrak{D} = \mathfrak{L}C^{-1}$$

in the paper's PT gauge, on an invertible sector.

- ▶ Main scope: §2.1–2.5 and the response bridge of Appendix B.
- ▶ Endpoint: the start of §3.1 through dressing, Eq. (3.9).
- ▶ §4's microscopic diffusion formula is **not derived in this talk**.

Conservation is exact; closure is the additional input

$$Q_i = \int dx \, q_i(x, t), \quad \partial_t q_i + \partial_x j_i = 0$$
$$\partial_t \bar{q}_i + \partial_x \bar{j}_i = 0, \quad \bar{q}_i = \langle q_i \rangle, \quad \bar{j}_i = \langle j_i \rangle.$$

Hydrodynamic assumption

After local relaxation, the slow conserved-density profiles determine local means. Locality and slow spatial variation permit a gradient expansion.

$$\bar{j}_i = F_i(\bar{q}) - \frac{1}{2} \mathfrak{D}_i^j(\bar{q}) \partial_x \bar{q}_j + O(\partial_x^2).$$

This is a relation between **expectations**, not $j_i = F_i(q)$ as an operator identity.

Source: Eqs. (2.1)–(2.4), (2.9).

What determines the Euler current and its slope?

$$\rho_\beta = Z^{-1} e^{-\sum_i \beta_i Q_i}, \quad \bar{q}_i(\beta) = \langle q_i \rangle_\beta, \quad \bar{j}_i(\beta) = \langle j_i \rangle_\beta.$$

Where $\bar{q}$ locally parametrizes the homogeneous GGE family,

$$F_i(\bar{q}) = \bar{j}_i(\beta(\bar{q})), \quad \boxed{A_i^j = \frac{\partial F_i}{\partial \bar{q}_j}}.$$

Physical meaning

One uniform state gives one point on the equation of state. Nearby uniform states determine its slope A . Spatial uniformity does not imply $A = 0$.

- A : thermodynamic response across states.
- $\partial_x \bar{q}$: spatial variation within a profile.

Source: Eqs. (2.3), (2.20).

The coefficient we want: the first correction beyond Euler

Substitute the mean-current closure into continuity:

$$\partial_t \bar{q}_i + \partial_x F_i(\bar{q}) = \frac{1}{2} \partial_x \left[\mathfrak{D}^j(\bar{q}) \partial_x \bar{q}_j \right].$$

Around a homogeneous stationary background,

$$\partial_t \delta \bar{q} = -A \partial_x \delta \bar{q} + \frac{1}{2} \mathfrak{D} \partial_x^2 \delta \bar{q}.$$

- ▶ First-gradient current $\Rightarrow$ second-gradient density evolution.
- ▶ For a nonlinear profile, the outer derivative also acts on $\mathfrak{D}(\bar{q})$.
- ▶ Paper convention: scalar $\nu \partial_x^2 \bar{q}$ has $\nu = \mathfrak{D}/2$.

Physical meaning

Writing the equation defines diffusion. Calculating $\mathfrak{D}$ makes it predictive.

Source: Eqs. (2.9)–(2.10); homogeneous linearization.

The fluctuations carry the transport information

In a homogeneous stationary state, preserve the displayed operator order:

$$S_{ij}(x, t) = \langle q_i(x, t) q_j(0, 0) \rangle^c, \quad \langle AB \rangle^c = \langle AB \rangle - \langle A \rangle \langle B \rangle.$$

$$M_n(t) = \int dx x^n S(x, t), \quad \boxed{M_0(t) = C.}$$

On a periodic ring of length ℓ ,

$$C_{ij} = \frac{1}{\ell} \langle Q_i Q_j \rangle^c.$$

- ▶ C : conserved fluctuation weight, not its spatial shape.
- ▶ For Hermitian commuting charges: $C = C^T \geq 0$.
- ▶ M_2 is an unnormalized correlation moment, not generally a probability variance.

Source: Eqs. (2.11)–(2.12); finite-volume covariance reconstruction.

Why the second moment turns density spreading into current noise

Define

$$G_{ij}(x, t) = \langle j_i(x, t) j_j(0, 0) \rangle^c, \quad K(t) = \int dx G(x, t).$$

Continuity at both insertions and translation invariance give

$$\partial_t^2 S = \partial_x^2 G.$$

Two spatial integrations by parts yield

$$M_2''(t) = \int dx x^2 \partial_x^2 G = 2 \int dx G = \boxed{2K(t)}.$$

Physical meaning

Current correlations measure the rate of growth of the density correlation's second moment. This is the microscopic–macroscopic bridge.

Boundary terms such as $x^2 \partial_x G$ and xG must vanish.

Source: Continuity derivation of the sum rule (2.13), Appendix A.

Ballistic memory must be subtracted before measuring diffusion

$$D = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T du K(u)$$

$$\mathfrak{L} = \lim_{T \rightarrow \infty} \int_{-T}^T du [K(u) - D]$$

Symbol	Definition	Role
D	Persistent current correlation	Ballistic t^2 term
$\mathfrak{L}$	Integrated residual correlation	Linear t term
$\mathfrak{D}$	Mean-current gradient coefficient	Constitutive diffusion

These are three different matrices. A finite residual integral is an assumption; slow tails can invalidate ordinary diffusion.

Source: Eqs. (2.14)–(2.16).

The exact time sum rule explains the powers of time

For $t > 0$, define $\mathcal{M}(t) = \frac{1}{2}[M_2(t) + M_2(-t)] - M_2(0)$.

$$\mathcal{M}(t) = \int_0^t ds \int_0^t ds' K(s - s') = \int_{-t}^t du (t - |u|) K(u).$$

$$K(u) = D + K_{\text{res}}(u) \implies \mathcal{M}(t) = Dt^2 + \mathfrak{L}t + o(t).$$

Physical meaning

The triangular weight counts time pairs with separation u . A persistent part fills the whole square (t^2); a finite-memory residual contributes along its diagonal (t).

No evenness assumption $K(u) = K(-u)$ is needed.

Source: Eqs. (2.13)–(2.16); finite-limit assumptions as in the notes.

How can a closure for means control a two-point function?

Response bridge used in Appendix B

Perturb the initial ensemble using a positive response coordinate η :

$$\frac{\delta \langle o(x, t) \rangle}{\delta \eta_j(y)} = \langle o(x, t) q_j(y, 0) \rangle^c.$$

Differentiate a mean relation $\Rightarrow$ obtain a relation for correlations.

- ▶ Commuting ensemble: numerator insertion minus normalization gives the connected covariance.
- ▶ General noncommuting quantum local sources: exponential differentiation gives a Kubo–Mori insertion, not automatically the ordered product.
- ▶ Here we retain the paper's ordered-response relation as an **input at the hydrodynamic order used**.

Source: Appendix B, Eq. (B.3); η avoids the GGE's negative β sign.

The chain rule gives the positive-time generator

At a uniform background, linearize the closure:

$$\frac{\delta \bar{j}_i(x, t)}{\delta \bar{q}_k(y, t)} = A_i^k \delta(x - y) - \frac{1}{2} \mathfrak{D}_i^k \partial_x \delta(x - y).$$

A variation of $\mathfrak{D}$ multiplies $\partial_x \bar{q} = 0$; the derivative of F remains.

$$\begin{aligned} J_{ij}(x, t) &:= \langle j_i(x, t) q_j(0, 0) \rangle^c \\ &= \int dy \frac{\delta \bar{j}_i(x, t)}{\delta \bar{q}_k(y, t)} S_{kj}(y, t) \\ &= A_i^k S_{kj} - \frac{1}{2} \mathfrak{D}_i^k \partial_x S_{kj}. \end{aligned}$$

$$\partial_t S = -A \partial_x S + \frac{1}{2} \mathfrak{D} \partial_x^2 S, \quad t > 0.$$

Source: Eqs. (B.1), (B.4), (2.19).

Negative time: evolve the later insertion, not a state backward

Define $H_{ij} = \langle q_i(x, t) j_j(0, 0) \rangle^c$. Continuity and clustering imply

$$\partial_x J = \partial_x H = -\partial_t S \quad \Rightarrow \quad J = H.$$

Translation preserves operator order:

$$H_{ij}(x, t) = \langle q_i(0, 0) j_j(-x, -t) \rangle^c.$$

For $t < 0$, the later current now carries the **second** index:

$$J_{ij} = S_{ik} A_j^k + \frac{1}{2} (\partial_x S_{ik}) \mathfrak{D}_j^k.$$

$$\partial_t S = -(\partial_x S) A^T - \frac{1}{2} (\partial_x^2 S) \mathfrak{D}^T, \quad t < 0.$$

The gradient sign uses $\partial_{(-x)} = -\partial_x$; the transpose records the index contraction.

Source: Appendix B, Eq. (B.5); negative branch of (2.19).

Moments isolate the coefficients without solving the profile

$$E := M_1(0), \quad M_0 = C, \quad M_1(t) = ACt + E, \quad \boxed{AC = CA^T}.$$

For $t > 0$,

$$\begin{aligned} \int dx \, x^2 \partial_x S &= -2M_1, & \int dx \, x^2 \partial_x^2 S &= 2C, \\ \frac{dM_2}{dt} &= 2AM_1 + \mathfrak{D}C. \end{aligned}$$

Integrate after substituting M_1 :

$$\boxed{N(t) := M_2(t) - M_2(0) = A^2 Ct^2 + (2AE + \mathfrak{D}C)t.}$$

Interpretation: Euler propagation gives t^2 ; spreading weight C and transporting the initial offset E both give linear terms.

Source: Eqs. (2.21)–(2.24); integrated form reconstructed in both notes.

Matching the two branches: what still obstructs $\mathfrak{L} = \mathfrak{D}C$?

For $t > 0$, integrate the negative-time branch to $-t$:

$$N(-t) = A^2 C t^2 + (C \mathfrak{D}^\top - 2EA^\top)t.$$

Compare $\frac{1}{2}[N(t) + N(-t)]$ with $Dt^2 + \mathfrak{L}t$:

$$D = A^2 C = A C A^\top$$

$$\mathfrak{L} = \frac{1}{2}(\mathfrak{D}C + C\mathfrak{D}^\top) + AE - EA^\top.$$

- ▶ E and the local-density convention still enter.
- ▶ The E factor follows direct integration; printed (2.26)–(2.27) have an *apparent* factor discrepancy (backup audit).
- ▶ This discrepancy does not affect the later $E = 0$ result.

Source: Eqs. (2.24)–(2.27), with the source-audit qualification.

The same total charge admits different local representatives

$$q'_i = q_i + \partial_x o_i, \quad j'_i = j_i - \partial_t o_i.$$

$$Q'_i = Q_i \quad (\text{vanishing boundary term}),$$
$$\partial_t q'_i + \partial_x j'_i = \partial_t q_i + \partial_x j_i + \partial_t \partial_x o_i - \partial_x \partial_t o_i = 0.$$

- ▶ This “gauge” is a choice of local conserved density, not electromagnetic gauge symmetry.
- ▶ All homogeneous stationary GGEs have the same density and current means: $F' = F$, $A' = A$.
- ▶ Total-charge covariance gives $C' = C$, hence $D' = D$.
- ▶ E and the constitutive coefficient $\mathfrak{D}$ can change at gradient order.

Source: Eq. (2.28), Appendix C.

Why $\mathfrak{D}$ can change while transport stays fixed

At leading order let $\bar{o} = O(\bar{q})$ and $B_i^j = \partial O_i / \partial \bar{q}_j$.

$$\bar{q}' = \bar{q} + B \partial_x \bar{q} + O(\partial_x^2), \quad -\partial_t \bar{o} = BA \partial_x \bar{q} + O(\partial_x^2).$$

The second relation uses Euler evolution $\partial_t \bar{q} = -A \partial_x \bar{q}$.

- ▶ Both the current improvement and rewriting $F(\bar{q})$ in $\bar{q}'$ change first-gradient terms.
- ▶ A total time derivative turns the integrated-current change into endpoint correlations.
- ▶ Hydrodynamic projection and appropriate relaxation make the endpoint differences cancel: $\mathfrak{L}' = \mathfrak{L}$.

Physical meaning

The observable transport weight $\mathfrak{L}$ is invariant under the stated assumptions; $\mathfrak{D}$ is a coefficient in a chosen set of local density variables.

Source: Discussion after (2.28), Appendix C.1; reconstruction in detailed note, branch D.

PT selects a compatible local-density representative

Use Θ for the paper's antiunitary PT involution (to reserve T for scattering).

$$\Theta H \Theta^{-1} = H, \quad \Theta Q_i \Theta^{-1} = Q_i, \quad \text{with locality and the paper's strong PT assumptions.}$$

Define the reflected partner at the same coordinates:

$$\tilde{q}_i(x, t) = \Theta q_i(-x, -t) \Theta^{-1}, \quad \tilde{q}_i - q_i = \partial_x b_i.$$

Average the representatives:

$$q_i^{\text{PT}} = \frac{1}{2}(q_i + \tilde{q}_i) = q_i + \partial_x(b_i/2).$$

Then $\Theta q_i^{\text{PT}}(x, t) \Theta^{-1} = q_i^{\text{PT}}(-x, -t)$; the accompanying current can be chosen likewise.

Source: Eqs. (2.29)–(2.31), Appendix C.2.

Antiunitarity: the reflection step needs a quantum qualification

For a PT-invariant GGE,

$$\langle \Theta O \Theta^{-1} \rangle = \langle O \rangle^*, \quad \Theta(AB)\Theta^{-1} = (\Theta A \Theta^{-1})(\Theta B \Theta^{-1}).$$

Thus the **ordered** correlator obeys

$$\boxed{S_{ij}(x, t)^* = S_{ij}(-x, -t)} \quad \Rightarrow \quad E = -E^*, \quad N(t)^* = N(-t).$$

Do not silently replace complex conjugation by equality

PT reflection alone does not establish $E = 0$ for arbitrary ordered quantum moments. Reality of the relevant moments is an additional step.

The paper states $E = 0$ and $N(t) = N(-t)$ in its PT gauge. We use these stated moment results, while keeping the reality/contact-term issue explicit.

Source: Ordered definition (2.11); paper's PT-gauge conclusions (2.32)–(2.33); note qualification.

The payoff: susceptibility-weighted diffusion

Accept the paper's PT-gauge moment statements:

$$E = 0, \quad N(t) = N(-t).$$

Matching the linear coefficients gives

$$\boxed{\mathfrak{D}C = C\mathfrak{D}^T}, \quad \boxed{D = A^2C, \quad \mathfrak{L} = \mathfrak{D}C.}$$

On an independent sector with invertible C ,

$$\boxed{\mathfrak{D} = \mathfrak{L}C^{-1}.}$$

- ▶ Inverse on the **right**; no arbitrary matrix commutation.
- ▶ $\mathfrak{D}C$ is symmetric; $\mathfrak{D}$ need not be.
- ▶ $C^{-1/2}\mathfrak{D}C^{1/2}$ is symmetric when $C > 0$.

Source: Eqs. (2.34)–(2.35).

A scalar packet checks both the factor C and the factor $1/2$

For $\gamma > 0$ and a centred point impulse of weight C ,

$$\partial_t S + v \partial_x S = \frac{\gamma}{2} \partial_x^2 S, \quad S(x, 0) = C \delta(x).$$

$$S(x, t) = \frac{C}{\sqrt{2\pi\gamma t}} \exp\left[-\frac{(x - vt)^2}{2\gamma t}\right], \quad t > 0.$$

$$\int S = C, \quad \frac{\int x S}{C} = vt, \quad \frac{\int (x - vt)^2 S}{C} = \gamma t.$$

$$\boxed{M_2 = Cv^2 t^2 + C\gamma t} \quad \Rightarrow \quad D = Cv^2, \quad \mathfrak{L} = C\gamma, \quad \mathfrak{D} = \gamma.$$

Physical meaning

Diffusion sets the broadening rate; susceptibility supplies the amount of fluctuation being broadened.

Source: Scalar teaching example in both notes; not a model-specific microscopic calculation.

A conserved current can be ballistic with zero Onsager part

If the exact local identity is $j_0 = q_1$, then

$$J_0 = \int dx j_0 = Q_1 \quad \text{is conserved.}$$

$$\begin{aligned} K_{0j}(s) &= \int dx \langle q_1(x, s) j_j(0, 0) \rangle^c \\ &= \langle Q_1 j_j(0, 0) \rangle^c =: k_j \quad (\text{time independent}). \end{aligned}$$

$$\boxed{D_{0j} = k_j, \quad \mathfrak{L}_{0j} = 0} \quad \Rightarrow \quad \boxed{\mathfrak{D}_0^i = 0} \quad (C \text{ invertible, PT gauge}).$$

- ▶ No residual Onsager weight does **not** mean no transport.
- ▶ A zero diffusion row does not freeze q_0 or remove damping from every coupled mode.

Source: Eqs. (2.36)–(2.38); the notes flag an apparent q_0/q_1 index issue in (2.38).

What must the integrable-model calculation now supply?

$$\mathfrak{D} = \mathfrak{L}C^{-1}, \quad \mathfrak{L} = \int_{-\infty}^{\infty} dt [K(t) - D]$$

when the stated transport limits exist.

- ▶ We have identified the coefficient, not evaluated an interacting model's $\mathfrak{L}$.
- ▶ Before computing a correlation, specify its stationary background.
- ▶ Bethe quasiparticles turn a many-charge GGE into rapidity-dependent occupation data.

$$\#\{\theta_a \in [\theta, \theta + d\theta]\} \simeq L\rho_p(\theta)d\theta.$$

ρ_p counts particles per physical length and rapidity; it is not normalized to one.

Source: Opening of §3, Eq. (3.1), and §3.1.

Occupied modes and available modes are different

For the initial one-species description with $p'(\theta) > 0$,

$$\rho_h = \rho_s - \rho_p, \quad \boxed{n(\theta) = \frac{\rho_p(\theta)}{\rho_s(\theta)}}.$$

Scattering changes the counting of available modes:

$$\boxed{\rho_s(\theta) = \frac{p'(\theta)}{2\pi} + \int d\alpha \, T(\theta, \alpha) \rho_p(\alpha)}.$$

$$T(\theta, \alpha) = \frac{1}{2\pi i} \partial_\theta \log S_{\text{scatt}}(\theta, \alpha).$$

- ▶ n : fraction of available Bethe modes occupied.
- ▶ Fermionic Bethe occupation has $0 \leq n \leq 1$; physical particles need not be fermions.
- ▶ T is now a scattering kernel, not the PT operator Θ .

Source: Eqs. (3.3)–(3.6); initially symmetric kernel as in (3.5).

Dressing is the self-consistent response to the occupied background

Insert $\rho_p = n\rho_s$ into the counting equation:

$$(1 - Tn)\rho_s = \frac{p'}{2\pi}.$$

The same structure defines dressing:

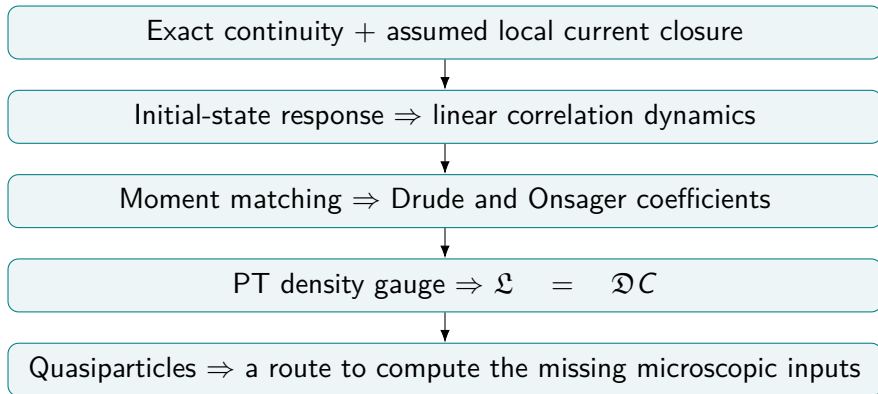
$$\boxed{h^{\text{dr}} = h + Tn h^{\text{dr}} = (1 - Tn)^{-1} h}, \quad \boxed{2\pi\rho_s = (p')^{\text{dr}}}.$$

$$(Tnh)(\theta) = \int d\alpha T(\theta, \alpha) n(\alpha) h(\alpha).$$

- ▶ $Tn \neq nT$ in general: the order of kernel and filling matters.
- ▶ $T = 0 \Rightarrow h^{\text{dr}} = h, \rho_s = p'/(2\pi)$.
- ▶ The inverse denotes a solvable integral equation, not an unconditional convergent geometric series.

Source: Eqs. (3.7)–(3.9), current reading endpoint.

The argument to take away



Conditions: local relaxation; long wavelengths; controlled boundary terms and transport limits; Appendix-B response input; projection for gauge invariance; PT moment statements with the quantum reality qualification.

Backup A: the signs in $\partial_t^2 S = \partial_x^2 G$

Using a second insertion at (y, s) ,

$$\begin{aligned}\partial_t S(x, t) &= -\partial_x \langle j_i(x, t) q_j(0, 0) \rangle^c, \\ \partial_t \langle j_i(x, t) q_j(0, 0) \rangle^c &= -\partial_s \langle j_i(x, t) q_j(0, s) \rangle^c \big|_{s=0} \\ &= \partial_y \langle j_i(x, t) j_j(y, 0) \rangle^c \big|_{y=0} \\ &= -\partial_x G(x, t).\end{aligned}$$

Therefore $\partial_t^2 S = \partial_x^2 G$ and

$$\int dx \, x^2 \partial_x^2 G = [x^2 \partial_x G]_{-\infty}^{\infty} - 2[xG]_{-\infty}^{\infty} + 2 \int dx \, G.$$

No operator interchange is used; sufficient spatial decay removes the boundaries.

Source: Continuity, stationarity, translation; Appendix A reconstruction.

Backup B: why the time-pair weight is triangular

Set $u = s - s'$, $v = s'$, with unit Jacobian. The square $0 \leq s, s' \leq t$ becomes

$$-t \leq u \leq t, \quad \max(0, -u) \leq v \leq \min(t, t - u).$$

The allowed v -interval has length $t - |u|$:

$$\int_0^t ds \int_0^t ds' K(s - s') = \int_{-t}^t du (t - |u|) K(u).$$

$$\int_{-t}^t (t - |u|) du = t^2, \quad \frac{1}{t} \int_{-t}^t (t - |u|) K_{\text{res}}(u) du \rightarrow \mathfrak{L}.$$

The last limit requires the stated convergence conditions (absolute integrability is sufficient). Neither stationarity nor the geometry assumes time-even K .

Source: Eq. (2.13); detailed note, supporting derivation B.

Backup C: why $AC = CA^T$, without local-operator swaps

Uniform GGE source derivatives with mutually commuting charges give

$$-\partial_{\beta_j} \bar{q}_k = C_{kj}, \quad \mathcal{B}_{ij} := -\partial_{\beta_j} \bar{j}_i = A_i^k C_{kj}.$$

Integrate $J_{ij} = H_{ij}$ at equal time:

$$\mathcal{B}_{ij} = \langle Q_i j_j(0) \rangle^c = \langle j_j(0) Q_i \rangle^c = \mathcal{B}_{ji}.$$

The middle equality uses $[\rho_\beta, Q_i] = 0$ and trace cyclicity, not $[Q_i, j_j] = 0$.

$$\boxed{AC = (AC)^T = CA^T}, \quad \hat{A} = C^{-1/2} AC^{1/2} = \hat{A}^T.$$

For density perturbations the corresponding metric is C^{-1} , not C .

Source: Eq. (2.23); detailed note, supporting derivation C.

Backup D: source differentiation and the connected subtraction

In a commuting source problem, define $W[\eta]$ and

$$\langle o \rangle_\eta = \frac{\text{Tr}(e^{W[\eta]} o)}{Z[\eta]}, \quad \frac{\delta W}{\delta \eta_j(y)} = q_j(y).$$

The quotient rule yields

$$\frac{\delta \langle o \rangle}{\delta \eta_j(y)} = \langle o q_j(y) \rangle - \langle o \rangle \langle q_j(y) \rangle.$$

For a general noncommuting variation,

$$\delta e^W = \int_0^1 d\lambda e^{(1-\lambda)W} (\delta W) e^{\lambda W}.$$

Thus a quantum exponential source normally gives a Kubo–Mori averaged insertion. The ordered response in (B.3) is a separately retained hydrodynamic input.

Source: Appendix B and the study-note response qualification.

Backup E: a local improvement changes the initial first moment

At equal time, define ordered correlations

$$R_{ij} = \langle o_i(x) q_j(0) \rangle^c, \quad U_{ij} = \langle q_i(x) o_j(0) \rangle^c, \quad V_{ij} = \langle o_i(x) o_j(0) \rangle^c.$$

Translation at the second insertion gives

$$S' = S + \partial_x R - \partial_x U - \partial_x^2 V.$$

Hence, under sufficient boundary decay,

$$\boxed{C' = C}, \quad \boxed{E' - E = - \int dx R + \int dx U}.$$

A derivative integrates to a boundary term for C , but its first moment leaves an integral:
 $\int x \partial_x R = - \int R.$

Source: Eq. (2.28); explicit reconstruction in both notes.

Backup F: Onsager invariance needs endpoint control

Use new symbols for the ordered integrated improvement correlations:

$$X_{ij}(u) = \int dx \langle o_i(x, u) j_j(0) \rangle^c,$$

$$Y_{ij}(u) = \int dx \langle j_i(x, u) o_j(0) \rangle^c,$$

$$Z_{ij}(u) = \int dx \langle o_i(x, u) o_j(0) \rangle^c.$$

Expanding $j'_i = j_i - \dot{o}_i$ and using stationarity gives

$$K' - K = -\dot{X} + \dot{Y} - \ddot{Z} = \dot{B}, \quad B = -X + Y - \dot{Z}.$$

$$\int_{-T}^T du (K' - K) = B(T) - B(-T).$$

If projection/relaxation gives $X(\pm\infty) = x$, $Y(\pm\infty) = y$, and $\dot{Z}(\pm\infty) = 0$, then $D' = D$ and $\mathfrak{L}' = \mathfrak{L}$.

Source: Appendix C.1; sufficient reconstruction in detailed note, supporting derivation D.

Backup G: projection keeps the conserved part, not a static local field

For integrated equal-time covariance $(a, b)_0 = \int dx \langle a(x)b(0) \rangle^c$,

$$\mathcal{P}a = (a, q_k)_0 (C^{-1})_{k\ell} q_\ell, \quad (a - \mathcal{P}a, q_j)_0 = 0.$$

The sufficient long-time projection statement is

$$\lim_{u \rightarrow \pm\infty} \int dx \langle a(x, u)b(0) \rangle^c = (a, q_k)_0 (C^{-1})_{k\ell} (q_\ell, b)_0.$$

- ▶ Spatial integration selects zero wave number: total charges are conserved.
- ▶ Local $q_i(x, t)$ still evolves by continuity.
- ▶ Spatial clustering is not temporal relaxation.
- ▶ Include all surviving conserved contributions; restrict C^{-1} to its nondegenerate sector.

Source: Hydrodynamic-projection input of Appendix C.1; detailed note, branch D.

Backup H: source audit points to keep separate from conclusions

1. **E-term normalization.** Integrating (2.24)–(2.25) gives

$$\frac{1}{2}[N(t) + N(-t)] = A^2 C t^2 + \left[\frac{1}{2}(\mathfrak{D}C + C\mathfrak{D}^\mathsf{T}) + AE - EA^\mathsf{T} \right] t.$$

Printed (2.26)–(2.27) give half the E contribution. This is an *apparent* discrepancy, not an author-confirmed erratum; it drops out at $E = 0$.

2. **Conserved-current index.** Under $j_0 = q_1$, use q_1 in the correlation proof; printed (2.38) displays q_0 .
3. **Quantum PT moments.** Reflection gives $E = -E^*$ and $N(t)^* = N(-t)$ directly. The stronger real-moment step remains qualified in our notes.

Source: Local paper extraction and the companion source-audit review.

Sources and scope

Primary source

J. De Nardis, D. Bernard, B. Doyon,
Diffusion in generalized hydrodynamics and quasiparticle scattering,
SciPost Phys. **6**, 049 (2019).
doi:10.21468/SciPostPhys.6.4.049; arXiv:1812.00767.

Local study artifacts used

minimum_note/GHD_minimum_note.tex (main argument)
notes/GHD_detailed_study_notes.md (derivations; §3.1 opening)
minimum_note/GHD_minimum_note_review.md (audit qualifications).

Coverage

§1–§2.5; Appendix B and selected Appendix C reconstruction;
§3.1 only through Eq. (3.9). Packet sketches and scalar checks are teaching illustrations. No new model-specific diffusion calculation is claimed.